Objective Type Questions

11. The number which is subtracted from $\frac{-6}{11}$ to get $\frac{8}{9}$ is _____.

(A) $\frac{34}{99}$ (B) $\frac{-142}{99}$ (C) $\frac{142}{99}$ (D) $\frac{-34}{99}$

12. Which of the following pairs is equivalent?

(A) $\frac{-20}{12}, \frac{5}{3}$ (B) $\frac{16}{-30}, \frac{-8}{15}$ (C) $\frac{-18}{36}, \frac{-20}{44}$ (D) $\frac{7}{-5}, \frac{-5}{7}$

13. $\frac{-5}{4}$ is a rational number which lies between _____ .

(A) 0 and $\frac{-5}{4}$ (B) –1 and 0 (C) –1 and –2 (D) –4 and –5

14. Which of the following rational numbers is the greatest?

(A) $\frac{-17}{24}$ (B) $\frac{-13}{16}$ (C) $\frac{7}{-8}$ (D) $\frac{-31}{32}$

15. The sum of the digits of the denominator in the simplest form of $\frac{112}{528}$ is _____ .

(A) 4 (B) 5 (C) 6 (D) 7

1.3 Basic Arithmetic Operations on Rational Numbers

All the rules and principles that govern fractions in the basic operations apply to rational numbers also.

1.3.1 Addition

There can be four different situations while doing addition.

Type 1 : Adding numbers that have same denominators

This is simply like adding like fractions and the result is the sum of the numerators divided by their common denominator.

Example 1.10

Add : $\frac{-6}{11}, \frac{8}{11}, \frac{-12}{11}$

Solution:

Write the given rational numbers in the standard form and then add them.

So, $\frac{-6}{11} + \frac{8}{11} + \frac{-12}{11} = \frac{-6+8-12}{11} = \frac{-10}{11}$

Type 2 : Adding numbers that have different denominators

After writing the given rational numbers in the standard form, use the LCM of their denominators to convert the numbers into equivalent rational numbers with a common denominator so that this reduces to Type1.

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